TATE RESOLUTIONS: FACTORIZATION THROUGH A DIVIDED POWER DGA QUASI-ISOMORPHISM

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Editorial note. 'Moreover, it is clear that H_m(A(m+1)) = 0 by construction' rests on the cohomology computations of the cited adjoining Examples (in-chapter template). Declared dependency: lemma-extend-differential is not carded (its source proof is 'a direct computation which is omitted'); recorded via external_references, not re-proved.

Allowed standard prerequisites (editorial).

- Differential extension to a DP variable. lemma-extend-differential (same chapter, named; NOT carded – build-gate exclusion). Extends d uniquely to A(m) after adjoining variables with prescribed boundaries; declared as an external reference, not re-proved (the source proof of that lemma is omitted).
- Adjoining DP variables, odd and even. example-adjoining-odd, example-adjoining-even (same chapter, named examples). Supply the divided power polynomial algebras in each degree and the cohomology bookkeeping behind 'H_m(A(m+1)) = 0 by construction'.

Source reference. Original definition 09PJ: sources/stacks-dpa/pinned-original-source.tex, source lines 1300–1300. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09PI: sources/stacks-dpa/pinned-original-source.tex, source lines 1241–1241. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09PH: sources/stacks-dpa/pinned-original-source.tex, source lines 1221–1221. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PM: sources/stacks-dpa/pinned-original-source.tex, source lines 1398–1398. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $R \to S$ be a homomorphism of commutative rings. There exists a factorization*

$$R \to A \to S$$

with the following properties:

(1) *(A, d, γ) is as in Definition 09PJ,*
(2) *$A \to S$ is a quasi-isomorphism (if we endow S with the zero differential),*
(3) *$A_0 = R[x_j : j \in J] \to S$ is any surjection of a polynomial ring onto S, and*
(4) *A is a graded divided power polynomial algebra over R.*

The last condition means that A is constructed out of A_0 by successively adjoining a set of variables T in each degree > 0 as in Example 09PH or 09PI. Moreover, if R is Noetherian and $R \to S$ is of finite type, then A can be taken to have only finitely many generators in each degree.

Proof. We write out the construction for the case that R is Noetherian and $R \to S$ is of finite type. Without those assumptions, the proof is the same, except that we have to use some set (possibly infinite) of generators in each degree.

Start of the construction: Let $A(0) = R[x_1, \ldots, x_n]$ be a (usual) polynomial ring and let $A(0) \to S$ be a surjection. As grading we take $A(0)_0 = A(0)$ and $A(0)_d = 0$ for $d \neq 0$. Thus $\mathrm{d} = 0$ and γ_n, $n > 0$, is zero as well.

Choose generators $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$ for the kernel of the given map $A(0) = R[x_1, \ldots, x_n] \to S$. We apply Example 09PH m times to get

$$A(1) = A(0)\langle T_1, \ldots, T_m \rangle$$

with $\deg(T_i) = 1$ as a graded divided power polynomial algebra. We set $\mathrm{d}(T_i) = f_i$. Since $A(1)$ is a divided power polynomial algebra over $A(0)$ and since $\mathrm{d}(f_i) = 0$ this extends uniquely to a differential on $A(1)$ by Lemma 09PM.

Induction hypothesis: Assume we are given factorizations

$$R \to A(0) \to A(1) \to \ldots \to A(m) \to S$$

where $A(0)$ and $A(1)$ are as above and each $R \to A(m') \to S$ for $2 \leq m' \leq m$ satisfies properties (1) and (4) of the statement of the lemma and (2) replaced by the condition that $H_i(A(m')) \to H_i(S)$ is an isomorphism for $m' > i \geq 0$. The base case is $m = 1$.

Induction step: Assume we have $R \to A(m) \to S$ as in the induction hypothesis. Consider the group $H_m(A(m))$. This is a module over $H_0(A(m)) = S$. In fact, it is a subquotient of $A(m)_m$ which is a finite type module over $A(m)_0 = R[x_1, \ldots, x_n]$. Thus we can pick finitely many elements

$$e_1, \ldots, e_t \in \mathrm{Ker}(\mathrm{d} : A(m)_m \to A(m)_{m-1})$$

which map to generators of this module. Applying Example 09PH or 09PI t times we get

$$A(m+1) = A(m)\langle T_1, \ldots, T_t \rangle$$

with $\deg(T_i) = m + 1$ as a graded divided power algebra. We set $\mathrm{d}(T_i) = e_i$. Since $A(m+1)$ is a divided power polynomial algebra over $A(m)$ and since $\mathrm{d}(e_i) = 0$ this extends uniquely to a differential on $A(m+1)$ compatible with the divided power structure. Since we've added only material in degree $m+1$ and higher we see that $H_i(A(m+1)) = H_i(A(m))$ for $i < m$. Moreover, it is clear that $H_m(A(m+1)) = 0$ by construction.

To finish the proof we observe that we have shown there exists a sequence of maps

$$R \to A(0) \to A(1) \to \ldots \to A(m) \to A(m+1) \to \ldots \to S$$

and to finish the proof we set $A = \operatorname{colim} A(m)$. □